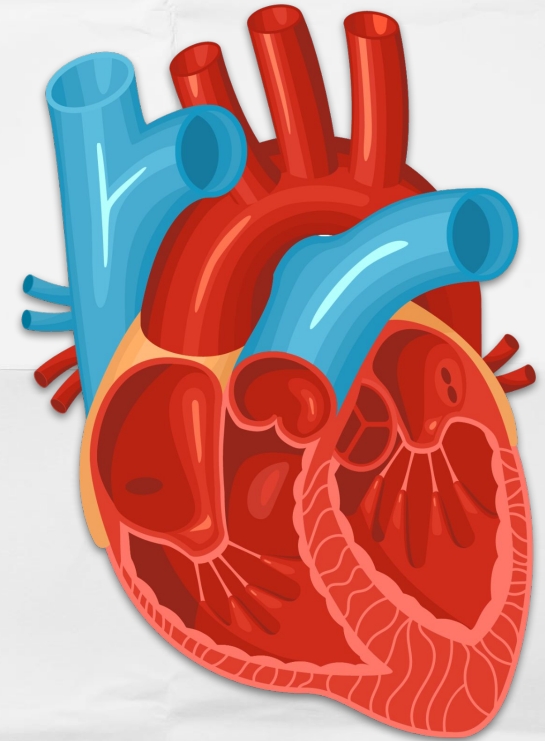


Automated Geometric Stenosis Quantification and CAD-RADS Support for Coronary Artery Disease Assessment and Patient Prioritization

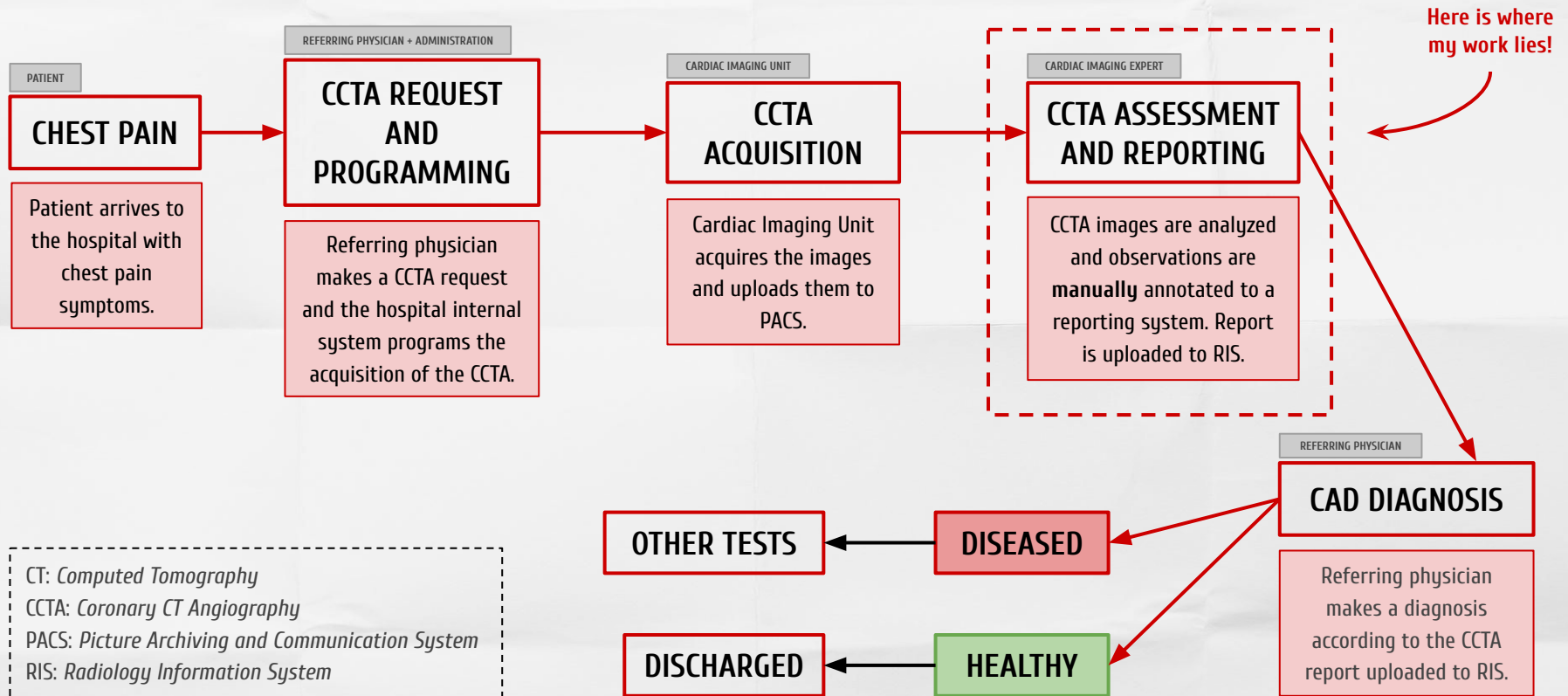
Adrià Cortés Cugat
Mathematical Engineering in Data Science

PhySense Update 4

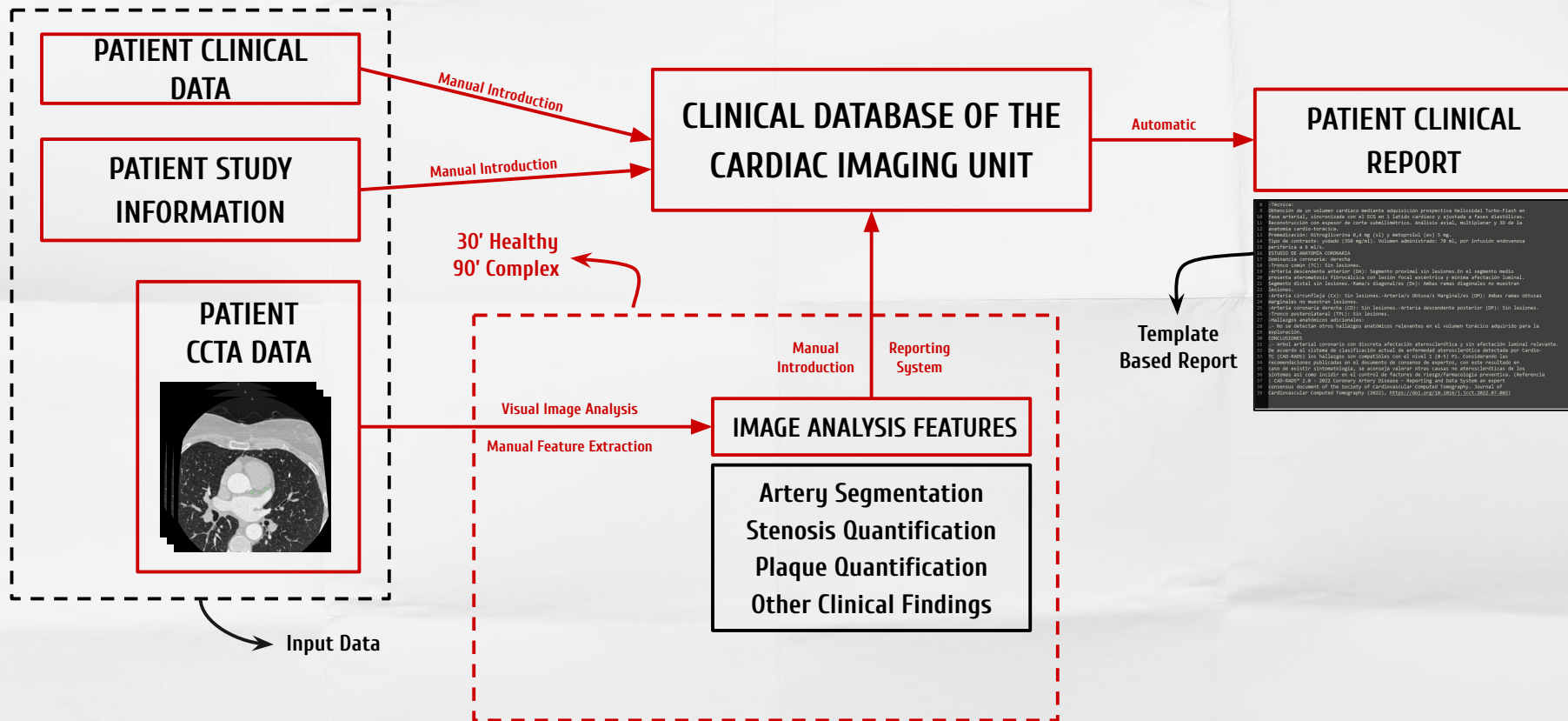


Note. From [Illustration of a human heart], in Ventricular Septal Defects Clinical Case Report Presentation [Google Slides template], by Slidesgo, n.d.
(<https://slidesgo.com/theme/ventricular-septal-defects-clinical-case-report>).

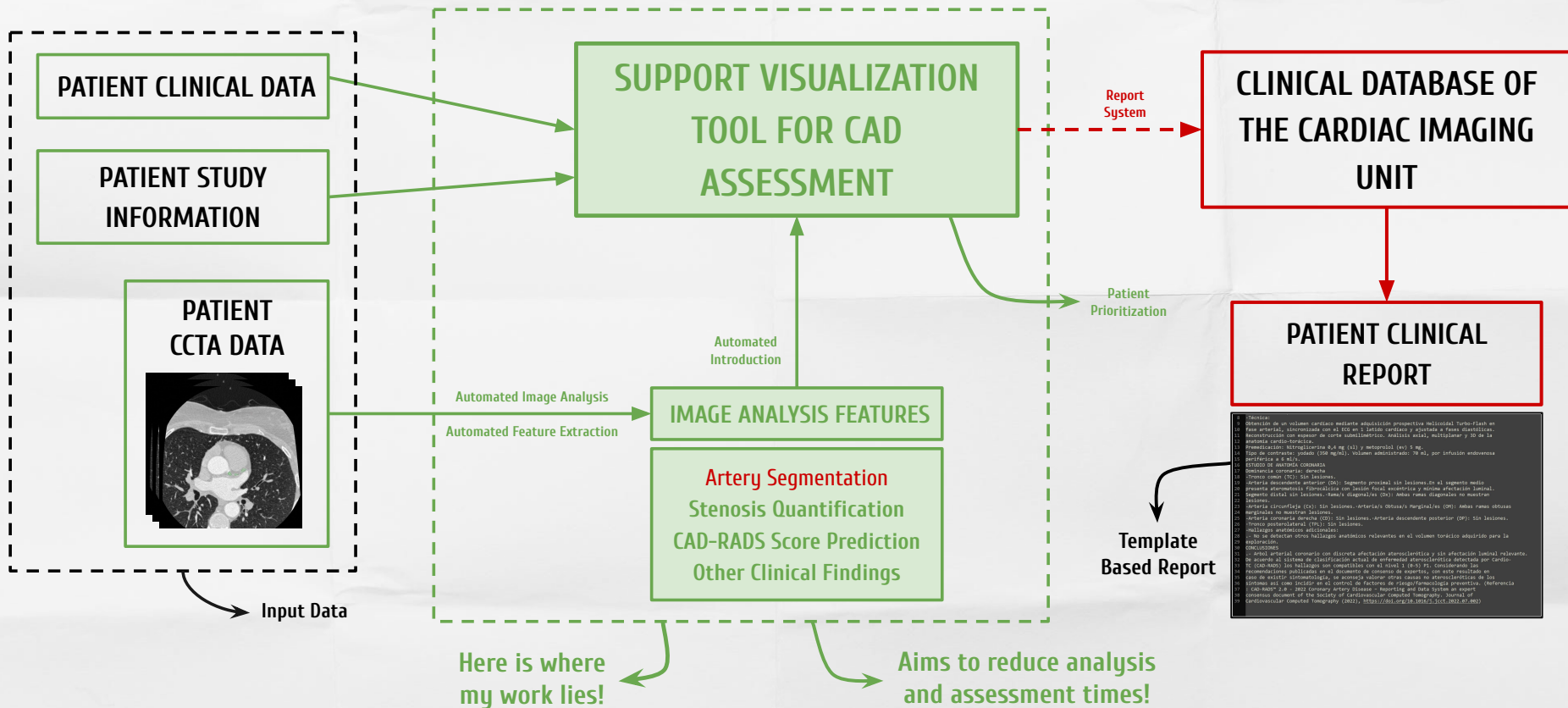
CURRENT CLINICAL WORKFLOW: PATIENT CAD DIAGNOSIS



CURRENT CLINICAL WORKFLOW: IMAGE ANALYSIS AND DIAGNOSIS

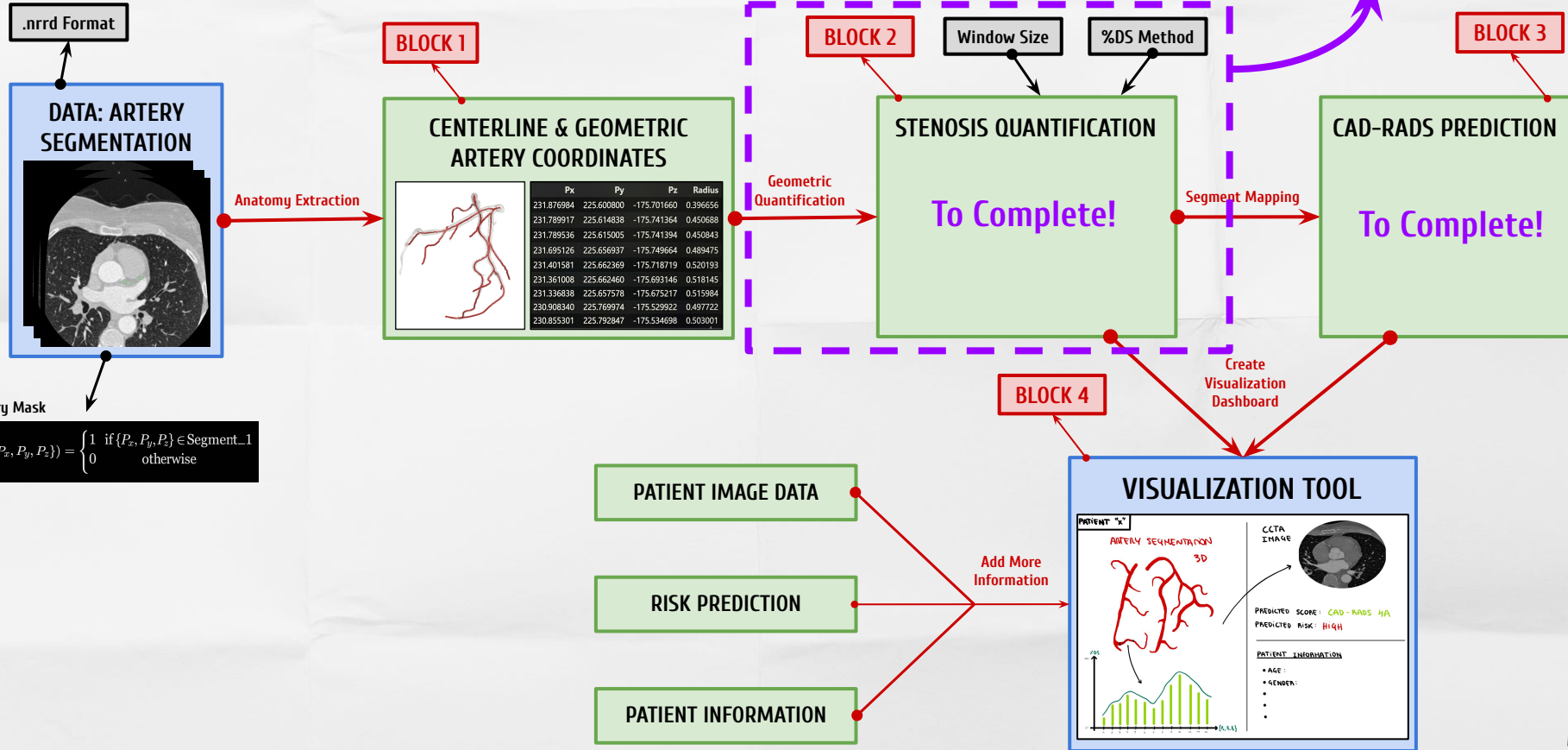


PROPOSED CLINICAL WORKFLOW: AUTOMATED SUPPORT FOR CAD ASSESSMENT



MASTER ARCHITECTURE: E2E AUTOMATED PIPELINE

Current Point!



BLOCK 1

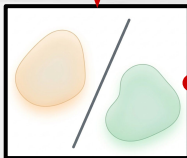
AUTOMATED ANATOMY EXTRACTION

Artery Seed Points

.nrrd Format



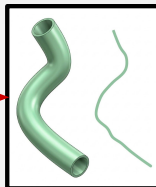
Load Binary Mask
Split RCA/LCA
Using Mass Center



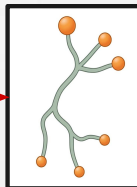
Input: Artery
Segmentation Data

For Each
Component

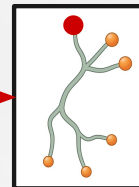
Voxel
Skeletonization



Endpoint
Detection



Ostium
Identification

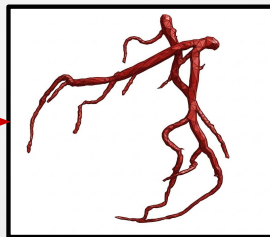


Point Coordinate
Conversion

Voxel $\rightarrow \{Px, Py, Pz\}$

Artery 3D Mesh Smoothed

Artery Mech Creation
& Smoothing



Centerline
Extraction

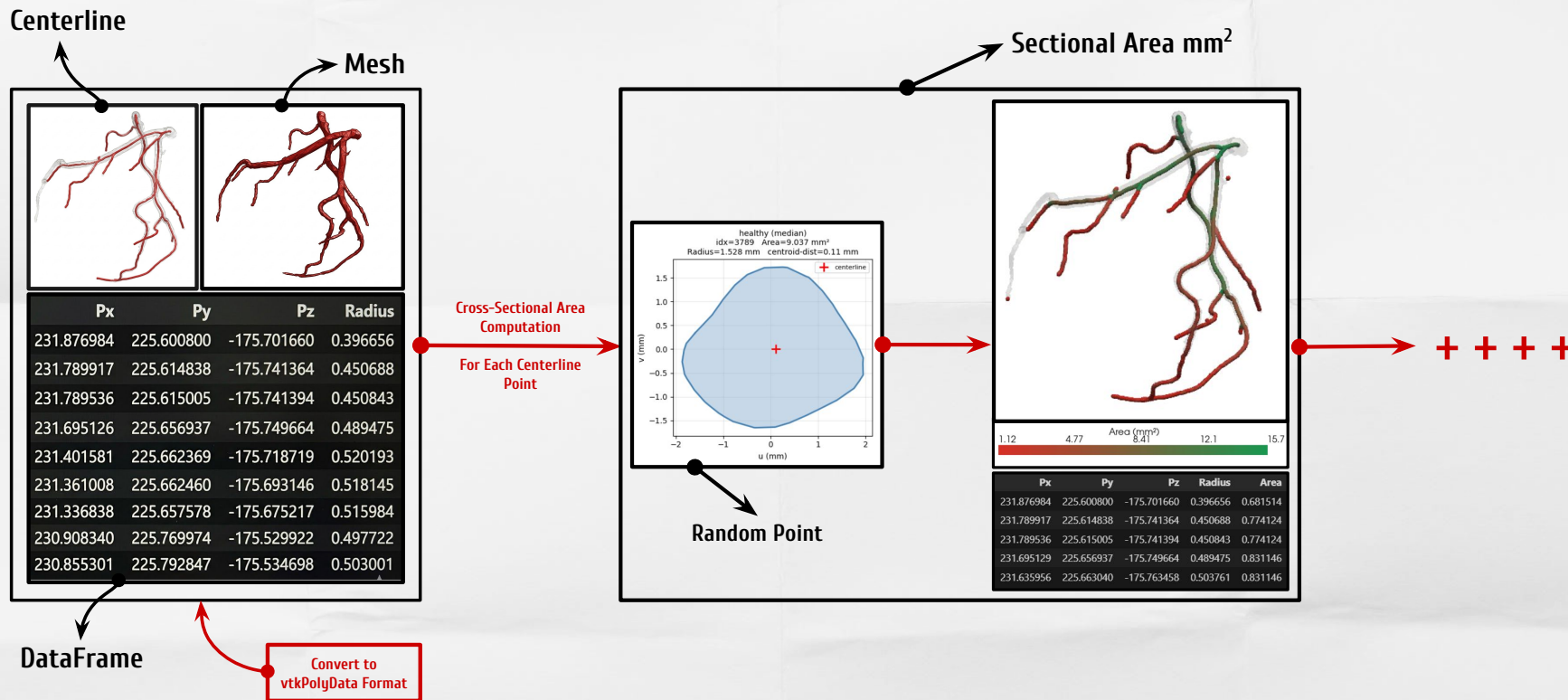


Artery Geometry
Data Frame

0	Sample_ID	Artery_Type	Px	Py	Pz	Radius
1	1	RCA	231.876984	225.600800	-175.701660	0.396656
2	1	RCA	231.789917	225.614838	-175.741364	0.450688
3	1	RCA	231.789536	225.615005	-175.741394	0.450843
4	1	RCA	231.695126	225.656937	-175.749664	0.489475
5	1	RCA	231.401581	225.662369	-175.718719	0.520193
6	1	RCA	231.361008	225.662460	-175.693146	0.518145
7	1	RCA	231.336838	225.657578	-175.675217	0.515984
8	1	RCA	230.908340	225.769974	-175.529922	0.497722
9	1	RCA	230.855301	225.792847	-175.534698	0.503001

BLOCK 2

GEOMETRIC STENOSIS QUANTIFICATION



Results: Normal_1.nrrd

```

———— Area Column — sanity —————
rows total      : 7,296
rows with NaN area: 0 (0.00%)
area min / max  : 0.061 / 72.034 mm2
area median     : 10.378 mm2
area quartiles  : Q1=3.069 Q3=12.330
  
```

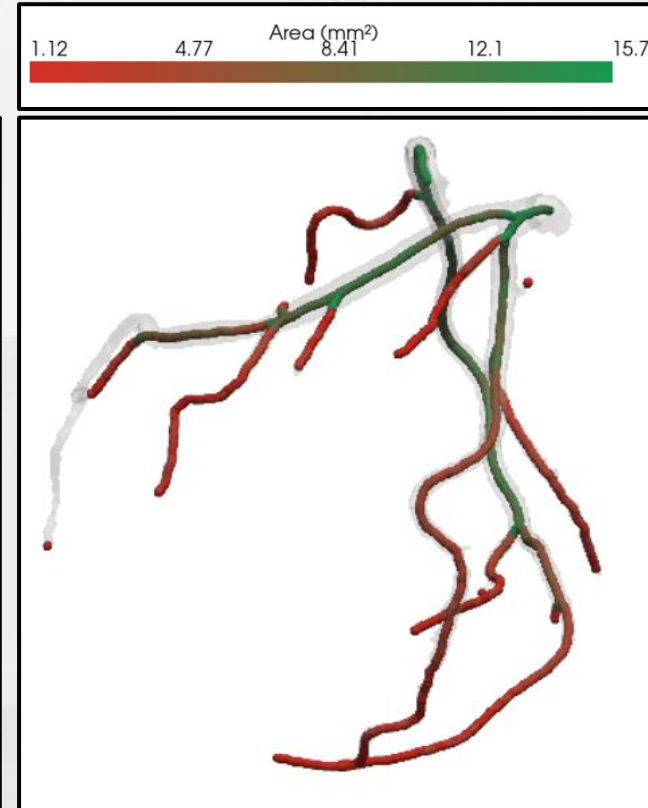
```

———— NaN Breakdown per Artery —————
[RCA] NaN rows: 0 / 3,990 (0.00%)
[LCA] NaN rows: 0 / 3,306 (0.00%)
  
```

```

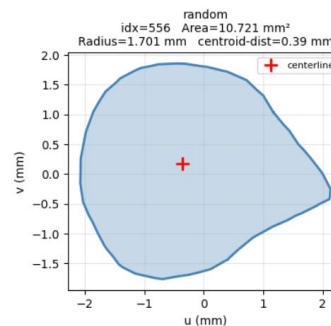
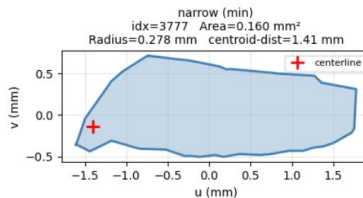
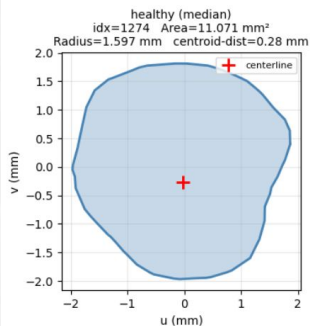
———— Radius ↔ Area correlation —————
[RCA] corr(Radius, Area) = +0.895 (expect strongly positive)
[LCA] corr(Radius, Area) = +0.745 (expect strongly positive)
  
```

Patient_ID	Artery_Type	Px	Py	Pz	Radius	Area
0	Normal_1	RCA	231.876984	225.600800	-175.701660	0.396656 0.681514
1	Normal_1	RCA	231.789917	225.614838	-175.741364	0.450688 0.774124
2	Normal_1	RCA	231.789536	225.615005	-175.741394	0.450843 0.774124
3	Normal_1	RCA	231.695129	225.656937	-175.749664	0.489475 0.831146
4	Normal_1	RCA	231.635956	225.663040	-175.763458	0.503761 0.831146

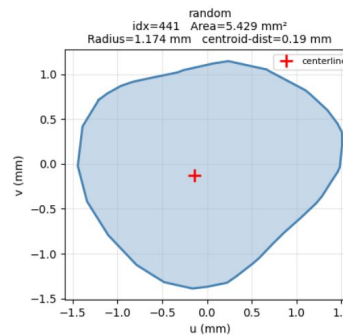
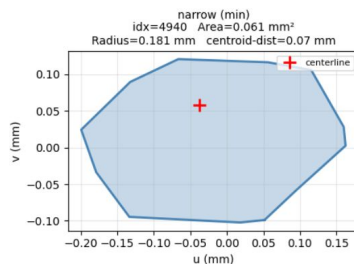
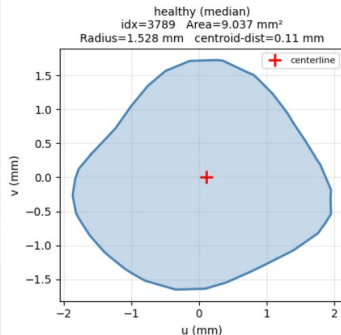


Results: Normal_1.nrrd

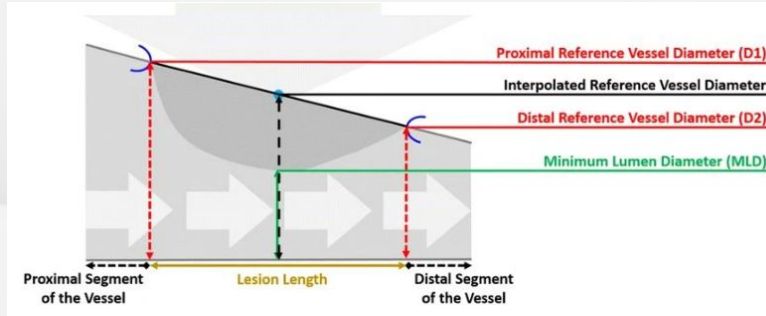
Normal_1 — RCA — Cross-section QA



Normal_1 — LCA — Cross-section QA



Next Steps



Sectional Area mm²

$$Stenosis(\%) = \left(1 - \frac{d_{min}}{\frac{d_{prox} + d_{dist}}{2}} \right) \times 100$$

Proximal/Distal Reference Values

1. Reference Value Computation → Value Methodology by Ela Burrull
2. %DS Computation For Each Point → Formula
3. Compute Maximum %DS Per Artery Segment
4. Block Validation and Start Block 3



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